

UNIVERSITY of **HOUSTON**
CENTER FOR ADVANCED COMPUTING & DATA SCIENCE



Machine Learning

Vistas in Advanced Computing

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Getting started

- Files with extension `.py` are programs for you to see.
- Files with extension `.csv` are data files to be used along with some of the `.py` programs.

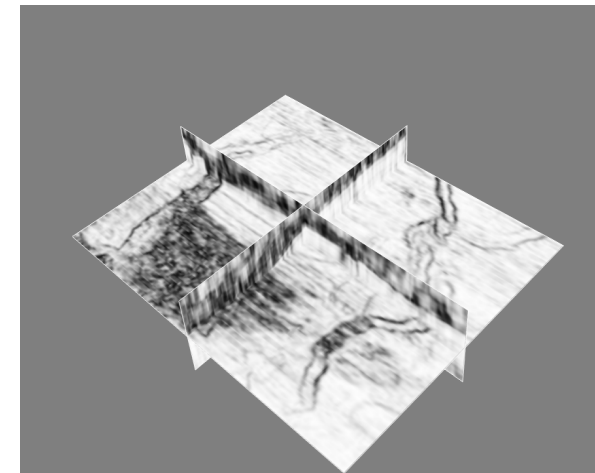
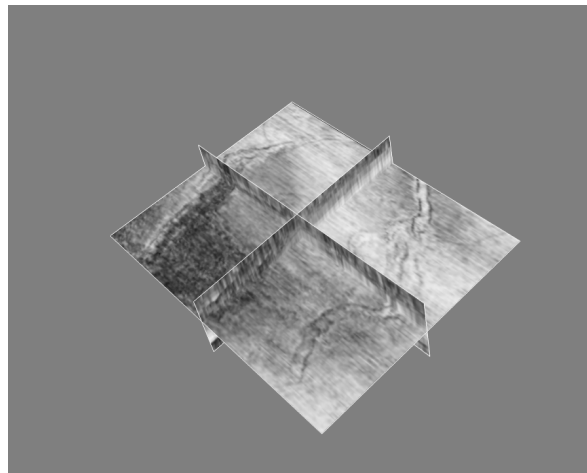
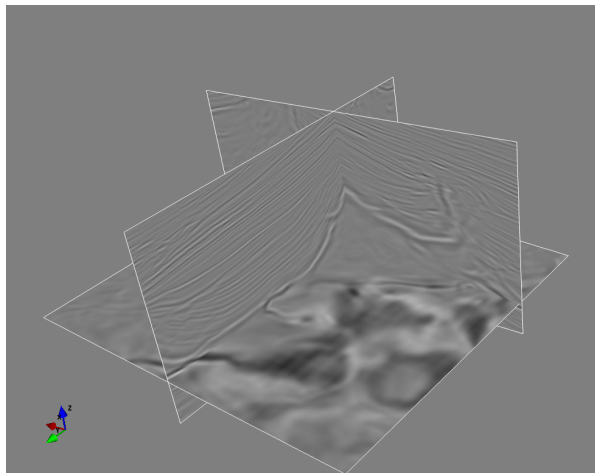
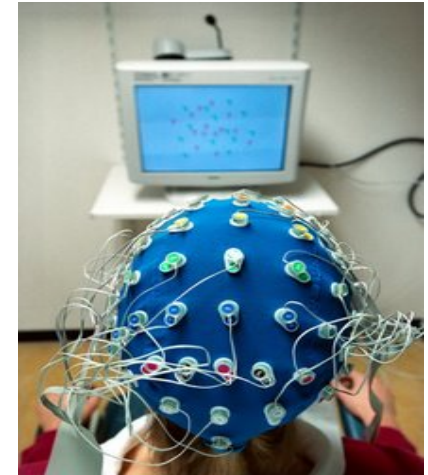
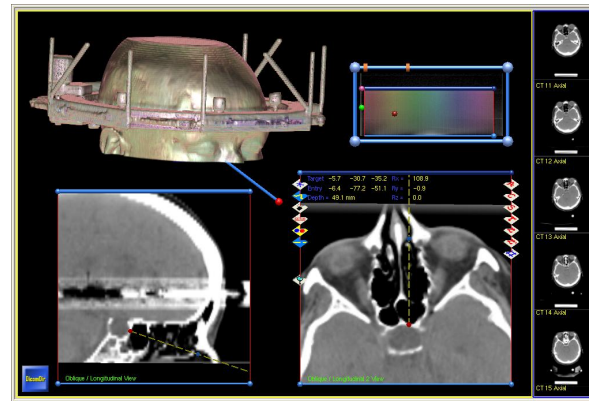
Overview

- Introduction to machine learning, unsupervised and supervised learning
- Introduction to machine learning methods and applications: KNN, Logistic Regression, Decision Tree, Random Forest, SVM, Neural Network
- Principal components analysis
- Clustering methods: Hierarchical, K-means
- Support Vector Machines
- Deep Learning algorithms: Multilayer Perceptrons, and Convolutional Neural Network

INTRODUCTION

In order to better model complex world real-data from different fields, including geosciences and medical sciences, one approach is to develop:

- Pattern recognition techniques
- Robust features
- Data Mining
- **Machine learning**
- Deep learning methods
- Predictive analytics



Machine Learning

- Data analytics:

Data analytics methodologies include exploratory data analysis, data mining, predictive analytics, and machine learning

Predictive Analytics: The process of identifying patterns in real-time and historical data to detect trends and determine future outcomes.

Supervised and unsupervised learning

- **Supervised learning** algorithms are trained with labeled data – in other words, data comprised of examples of the answers wanted. Most machine learning is supervised.
- **Unsupervised learning** is used on data with no labels, and the goal is to find relationships in the data.

Model training and evaluation: A machine learning model is an abstraction of the question you are trying to answer or the outcome you want to predict. Models are trained and evaluated from existing data.

Training data

- When you train a model from data, you use a known data set and make adjustments to the model based on the data characteristics to get the most accurate answer. In Machine Learning, a model is built from an algorithm that processes training data and functional modules, such as a scoring module.
- In supervised learning, if you are training a model, e.g. binary, you use a set of data that are labeled as either Class 1 or Class 2. You split your data set randomly, and use part to train the model and part to test or evaluate the model.

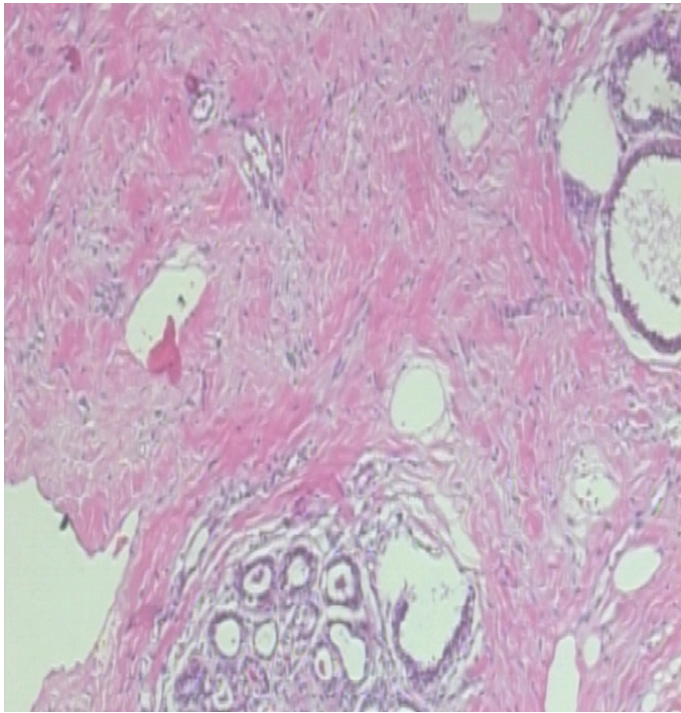
Evaluation data

- Once you have a trained model, evaluate the model using the remaining test data. You use data you already know the outcomes for, so that you can tell whether your model predicts accurately.

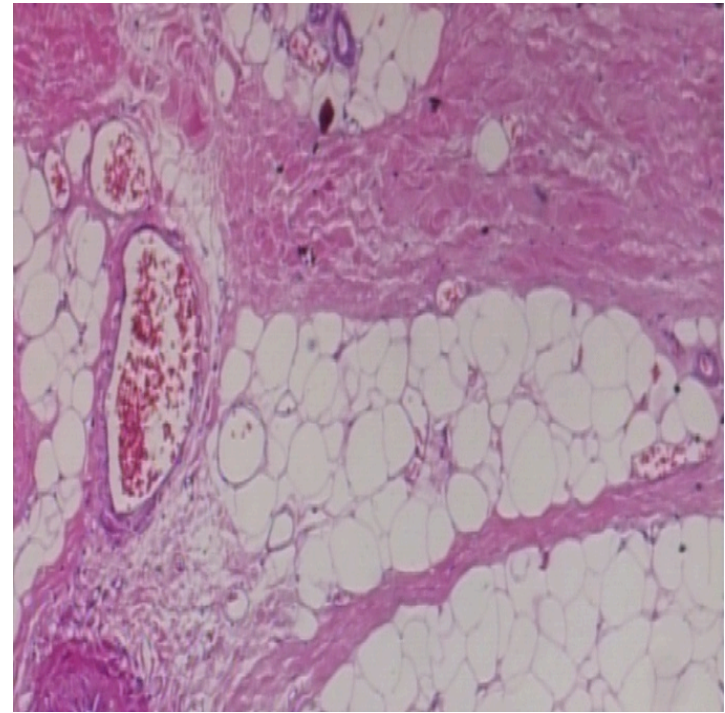
Digital Pathology: Breast Cancer

Table I. Number of Images

Images	Benign	Malignant
Total	370	880



(a)



(b)

Fig. 1. (a) A slide of breast benign tumor, and (b) a slide of breast malignant tumor.

- In our methodology, we training a CNN on a set of randomly selected patch, 80% of the dataset was used to training and 20% was used as the testing set. Table I reports the results with several metrics of the CNN architecture in the subset of images under analysis. We can really see that our classifier is achieving a 91% of precision on average in the classification.

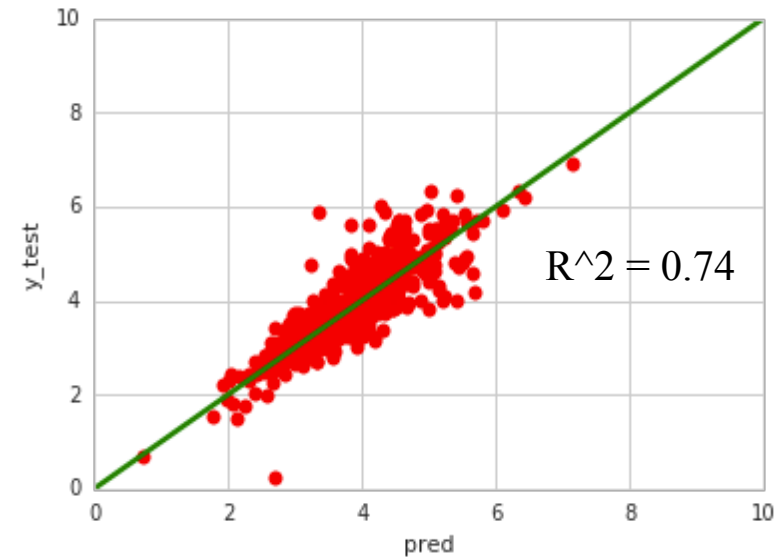
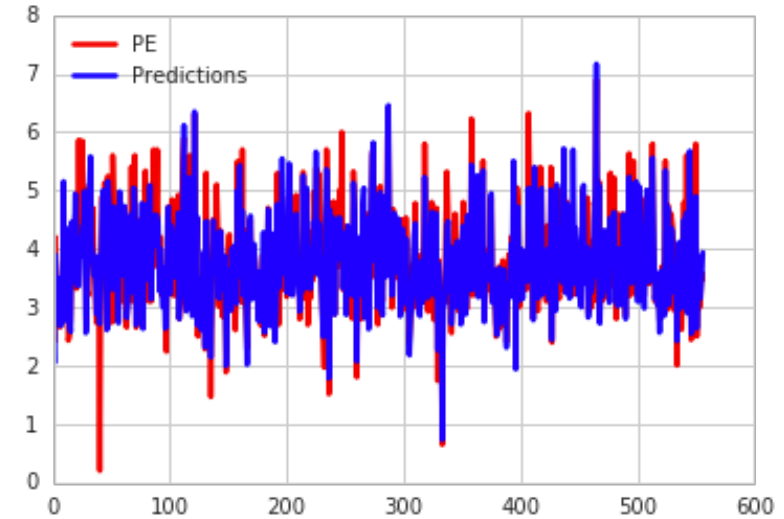
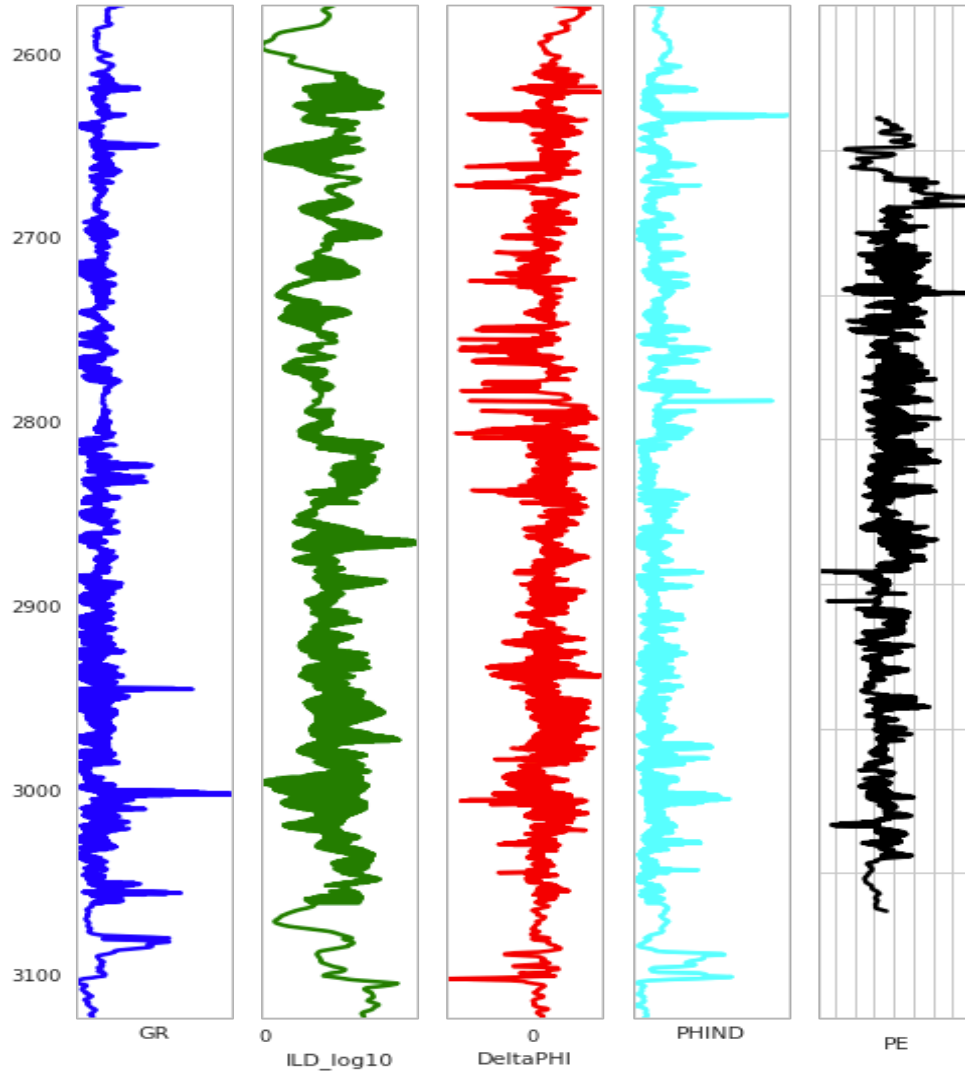
Table I. Summary of best results

class	precision	recall	f1-score	support
Benign	0.852417	0.865856	0.859084	2220
Malignant	0.943222	0.936970	0.940086	5280
avg/total	0.916344	0.915920	0.916109	7500

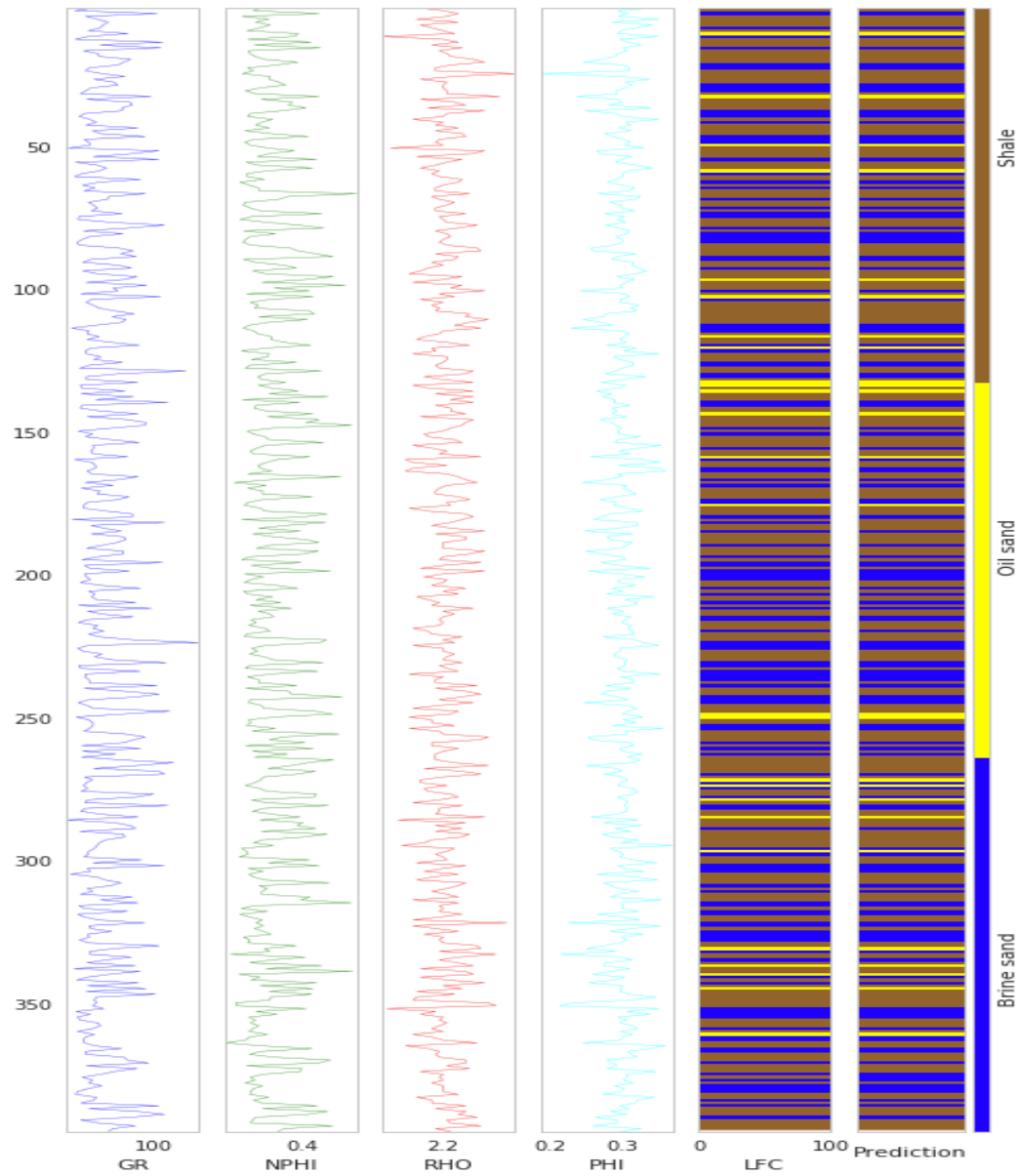
Precision and recall are metrics that give more insights into how the classifier performs for individual images. Precision is the probability that given a classification result for a sample, the sample actually belongs to that class. Recall is the probability that a sample will be correctly classified for a given class. The F1 score combines both to give a single measure of relevancy of the classifier results. Practically, as shown in the relative statistics for each class, the tow classes had high rates of accuracy.

Deep Learning

Automatic Identification of Formation lithology from Well Log Data: A Deep Learning Approach



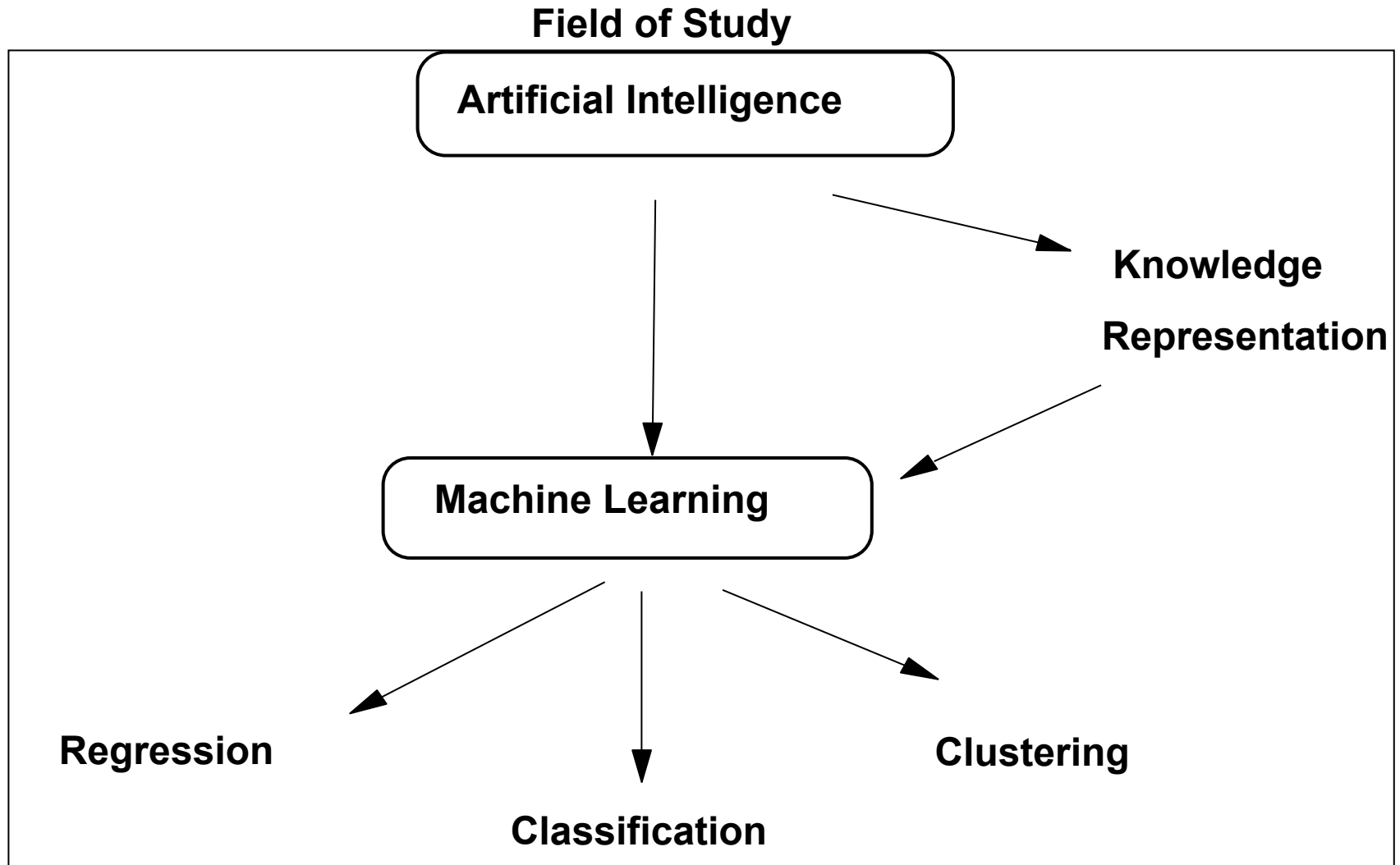
Optimized LFC classification accuracy = 0.99



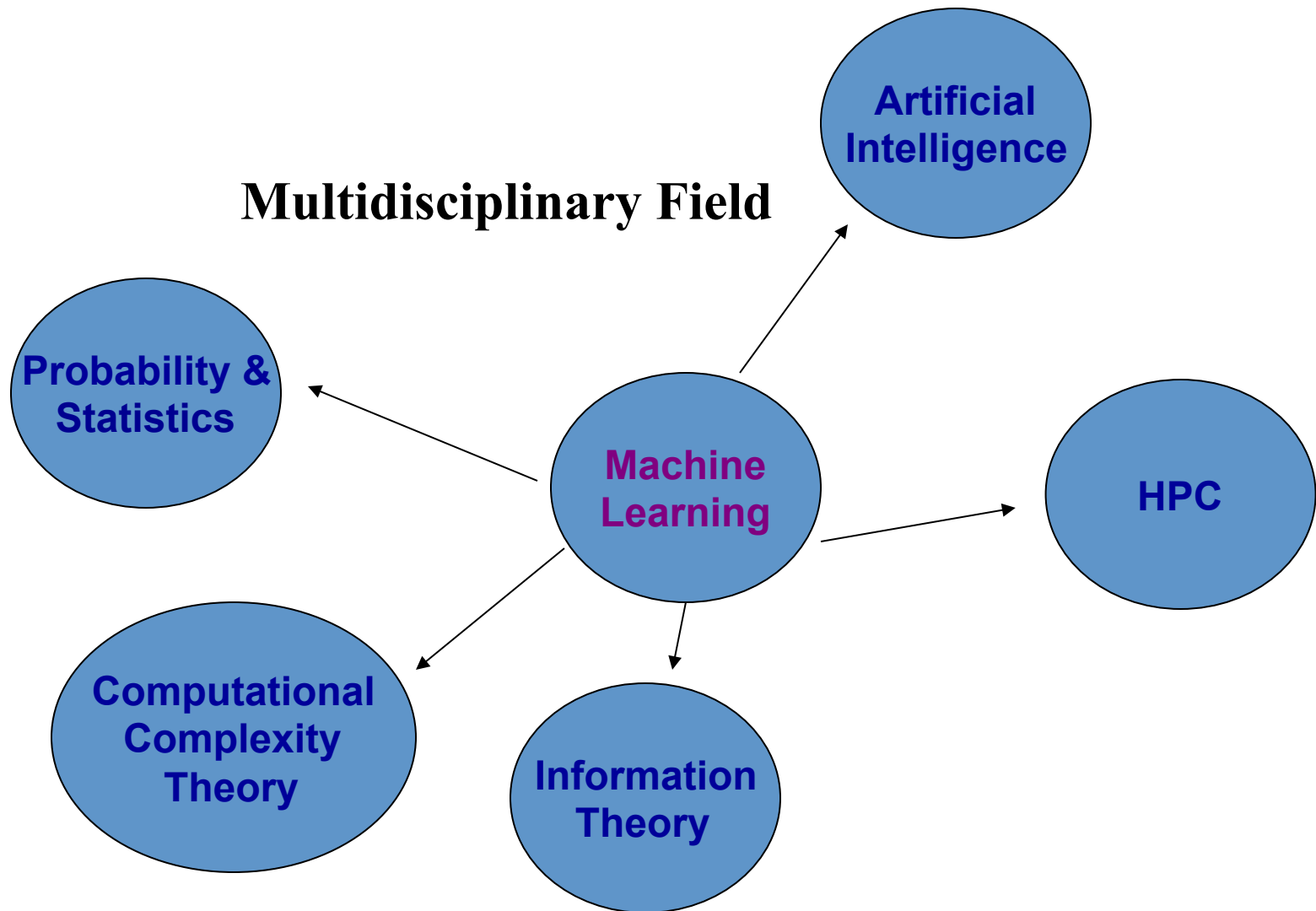
Machine Learning

- Where does machine learning fit in data analytics?
- What is machine learning?
- Where can machine learning be applied

Artificial Intelligence: The ability of a computer or machine to act with human-like intelligence, making the same sort of decisions and applying the same sort of reasoning that a human being would.

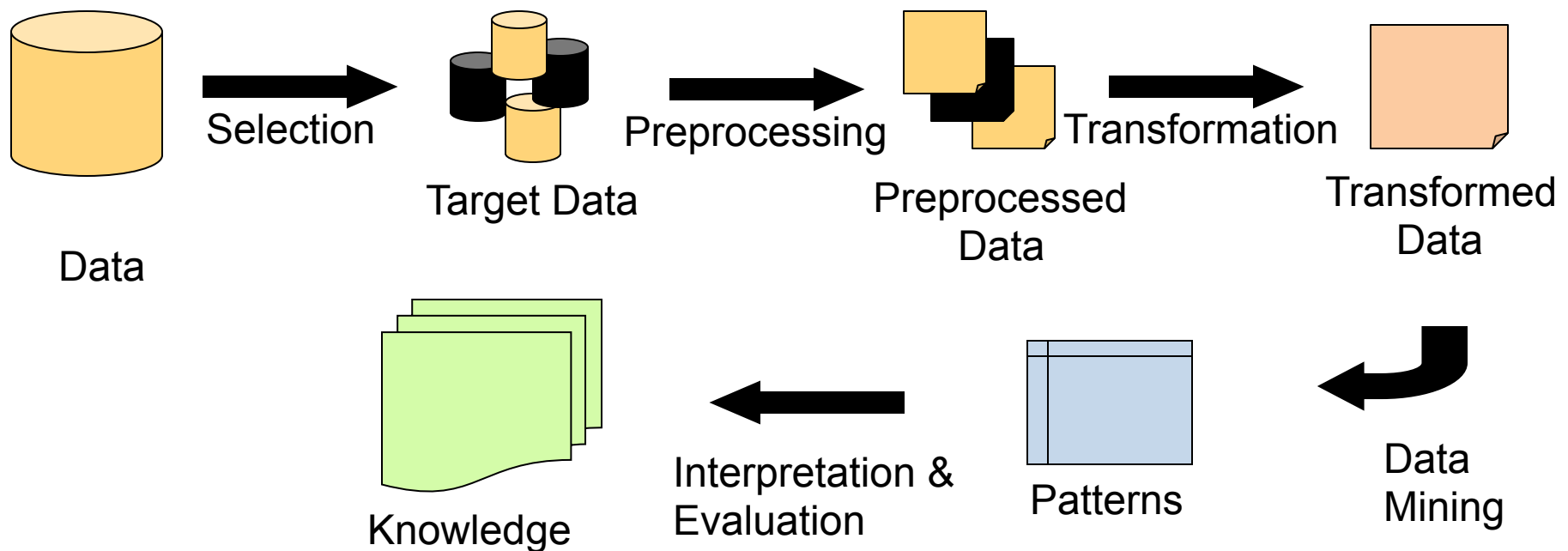


Where does machine learning fit in data analytics?



Where does machine learning fit in computer science?

Knowledge Discovery and Data Mining



Machine Learning

Where does machine learning fit in data analytics?

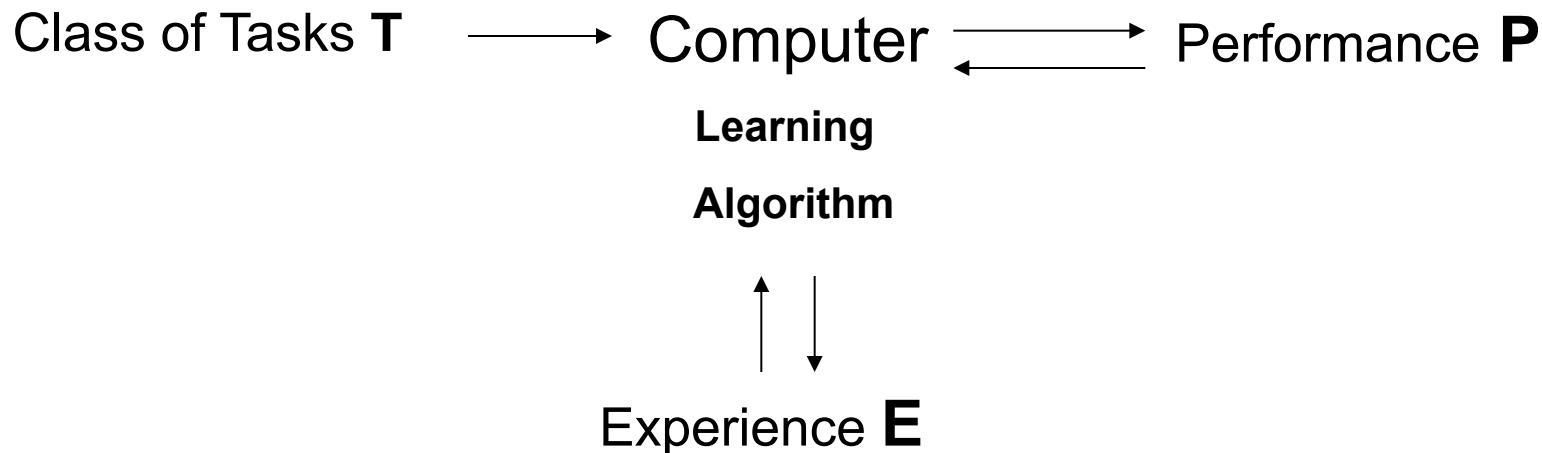
What is machine learning?

Where can machine learning be applied?

What is Machine Learning?

Definition

Machine learning is the study of how to make computers learn or adapt; the goal is to make computers improve their performance through experience.

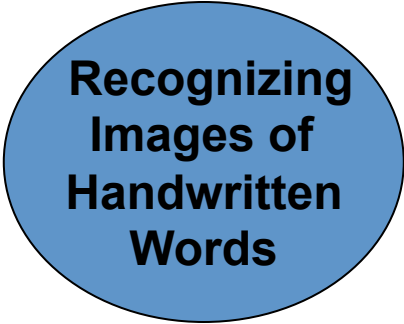


What is Machine Learning?

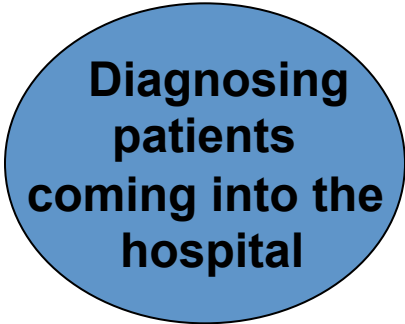
Definition

Class of Tasks:

It is the kind of activity on which the computer will learn to improve its performance. Examples:



**Recognizing
Images of
Handwritten
Words**



**Diagnosing
patients
coming into the
hospital**

What is Machine Learning?

Definition

Experience and Performance

Experience: What has been recorded in the past

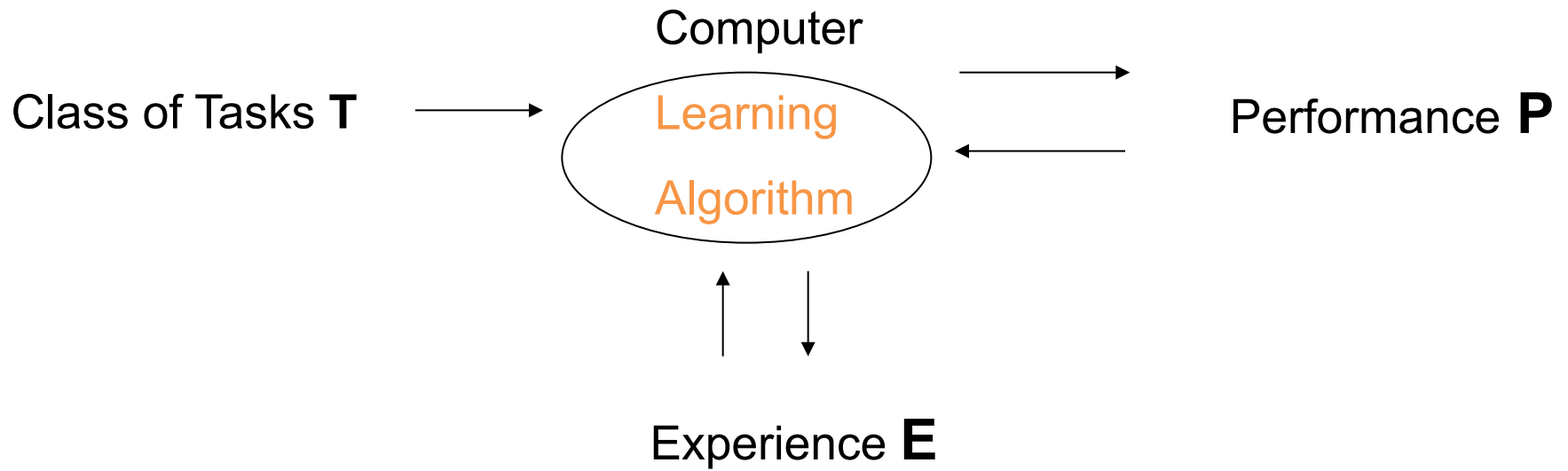
Performance: A measure of the quality of the response or action.

Experience: a database of handwritten images with their correct classification

Performance: Accuracy in classifications

What is Machine Learning?

Definition



What is Machine Learning?

Definition

Designing a Learning System

The Representation of the Target Knowledge

Example: Diagnosing a patient coming into the hospital.

- Features:
- X_1 : Temperature
- X_2 : Blood pressure
- X_3 : Blood type
- X_4 : Age
- X_5 : Weight
- Etc.

Given a new example $X = \langle x_1, x_2, \dots, x_n \rangle$

$$F(X) = w_1x_1 + w_2x_2 + w_3x_3 = \dots + w_nx_n$$

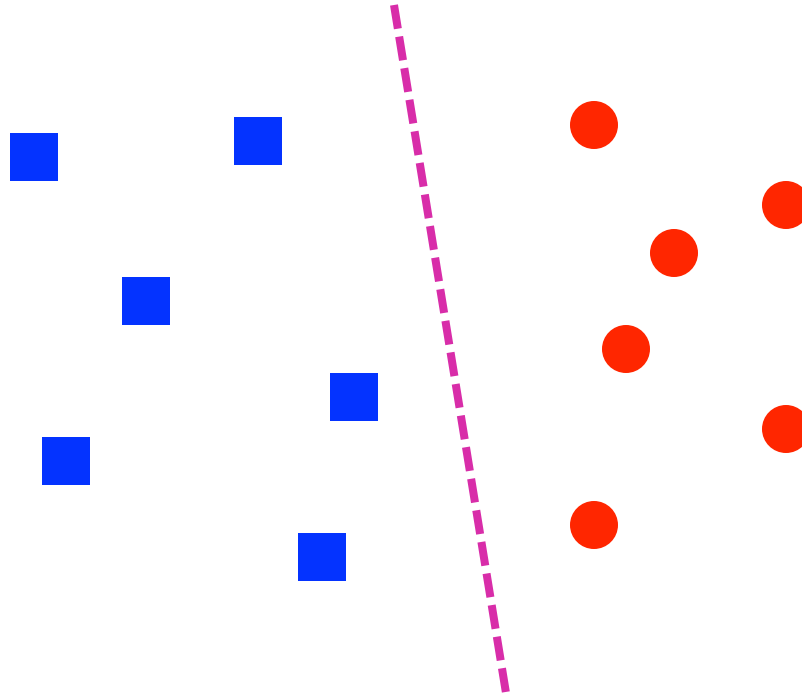
If $F(X) > T$ predict heart disease
otherwise predict no heart disease

What is Machine Learning?

Definition: is a form of induction

- Given examples of a function $(x, f(x))$
 - - Supervised learning (because we're given $f(x)$)
 - - Don't explicitly know f
 - Rather, trying to learn f from the data
 - - Labeled data set (i.e., the $f(x)$'s)
 - - Training set may be noisy, e.g., $(x, (f(x) + \varepsilon))$
 - - Notation: $(x_i, f(x_i))$ denoted $(x^{(i)}, y^{(i)})$
 - - $y^{(i)}$ sometimes called t_i (t for “target”)
- Predict function $f(x)$ for new examples x
 - - Discrimination/Prediction (Regression): $f(x)$ continuous
 - - Classification: $f(x)$ discrete
 - - Estimation: $f(x) = P(Y = c|x)$ for some class c

Example: Classifiers: Linear



- Find a *linear function* to separate the classes:

$$f(\mathbf{x}) = \text{sgn}(\mathbf{w} \cdot \mathbf{x} + b)$$

TIME CODE

- Simple example to understand machine learning
- Program: Example1_ML.py

Machine Learning

Where does machine learning fit in data analytics?

What is machine learning?

- Definition
- Types of Machine Learning
- Where can machine learning be applied?

What is Machine Learning?

Types of Machine Learning

Supervised Learning

- Each example or object has a class attached to it.
- We try to learn a mapping from examples to classes.

Two modes: classification and regression

Machine learning algorithms abound:

- Decision Trees
- Rule-based systems
- Neural networks
- Nearest-neighbor
- Support-Vector Machines
- Deep learning

What is Machine Learning?

Types of Machine Learning

- Supervised Learning
- Unsupervised Learning

What is Machine Learning?

Types of Machine Learning

Supervised Learning

- The set of data (training data) consists of a set of input data and correct responses corresponding to every piece of data
- Based on this training data, the algorithm has to **generalize** such that it is able to correctly (or with a low margin of error) respond to all possible inputs
- **In essence:** The algorithm should produce sensible outputs for inputs that were not encountered during training
- Also called learning from exemplars

What is Machine Learning?

Types of Machine Learning

Supervised Learning: Classification Problems

- Consists of taking input vector and deciding which of the N classes they belong to, based on training from exemplars of each class
- Is discrete: an example belongs to precisely one class, and the set of classes covers the whole possible output space
- **How it is done:** Find **decision boundaries** that can be used to separate out the different classes

What is Machine Learning?

Types of Machine Learning

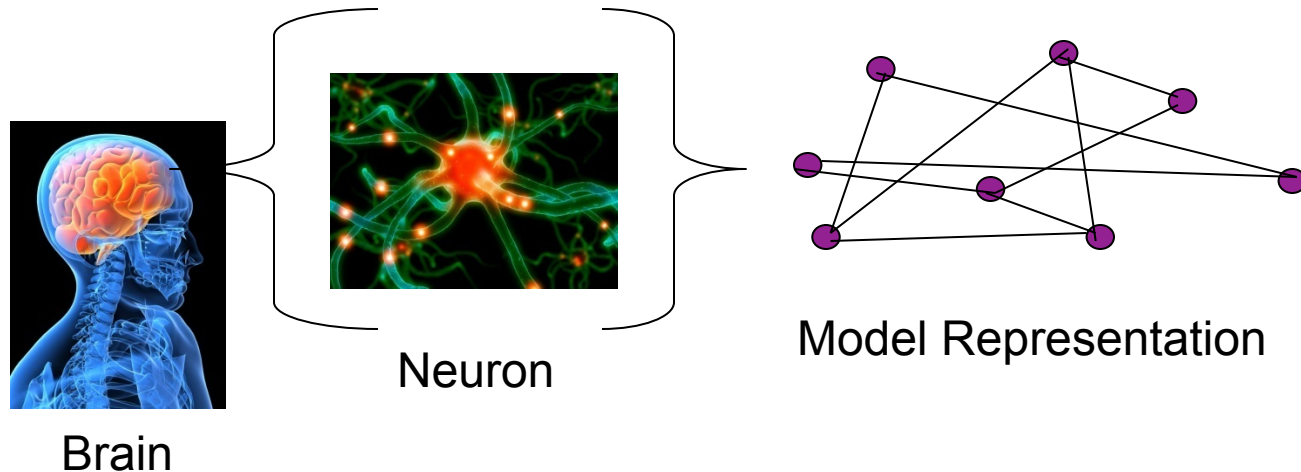
Supervised Learning: Regression Problems

- Given some data, you assume that those values come from some sort of function and try to find out what the function is
- **In essence:** You try to fit a **mathematical function** that describes a curve, such that the curve passes as close as possible to **all** the data points
- So, regression is essentially a problem of **function approximation**

What is Machine Learning?

Types of Machine Learning

Supervised Learning – Deep Learning



Artificial Neural Networks are crude attempts to model the highly massive parallel and distributed processing we believe takes place in the brain.

Consider:

- the speed at which the brain recognizes images;
- the many neurons populating a brain;
- the speed at which a single neuron transmits signals.

What is Machine Learning?

Types of Machine Learning

Unsupervised Learning

- Conceptually different problem
- No information about correct outputs are available
- Classification?
- No information about the correct classes. But if we design our algorithm so that it exploits similarities between inputs so as to cluster inputs that are similar together, this might perform classification automatically
- **In essence:** The aim of unsupervised learning is to find clusters of similar inputs in the data **without being explicitly told** that some data points belong to one class and the other in other classes. The algorithm has to discover this similarity by itself

Machine Learning

Where does machine learning fit in data analytics?

What is machine learning?

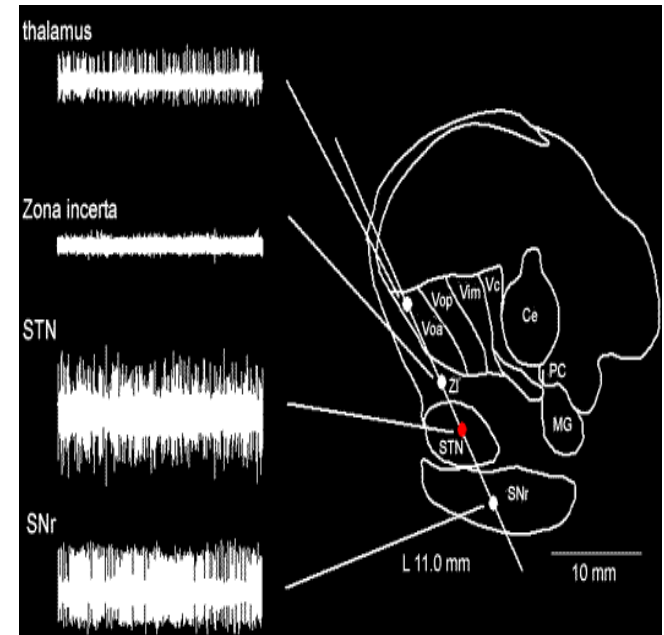
Where can machine learning be applied?

Where can machine learning be applied?

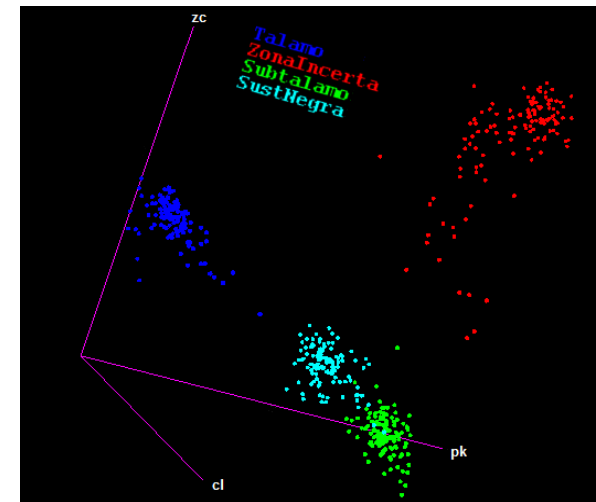
Application 1

Learning to classify subcortical structures for DBS

■ Features



Machine learning



Where can machine learning be applied?

Application 1

Class of Tasks: Learning to recognize subcortical structures during DBS from signals

Knowledge: Features that describe each subcortical structure

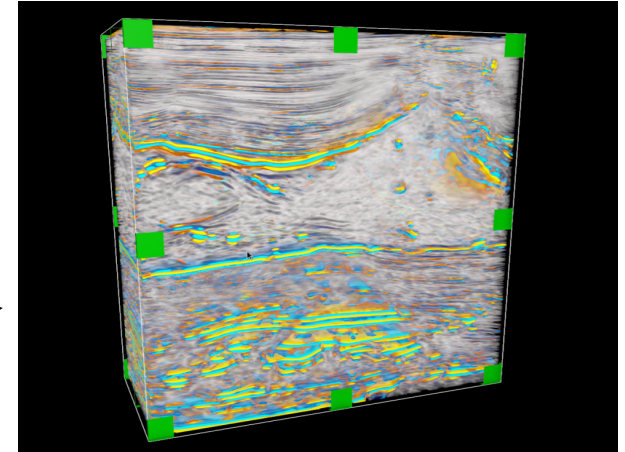
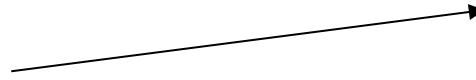
Performance Module: Accuracy in classification

Where can machine learning be applied?

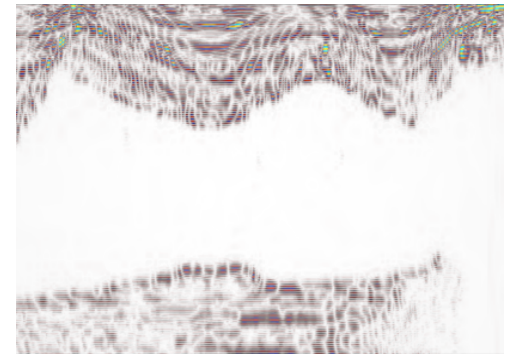
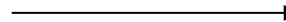
Application 2

Learning to classify salt structures in seismic data for reservoir (oil/gas) characterization

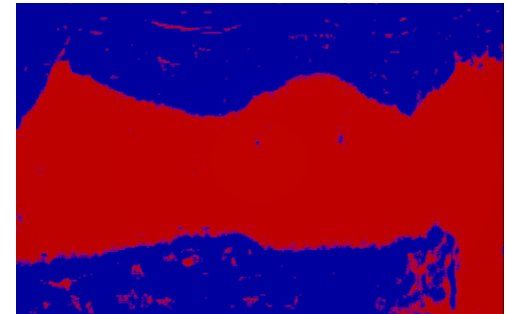
- 3D Seismic data



- Features



Machine
learning
(0 salt, red, 1
sediments, blue)



Where can machine learning be applied?

Application 2

Class of Tasks: Learning to classify geological structures

Knowledge: database of images with seismic attributes and correct classification

Performance Module: Accuracy in classification

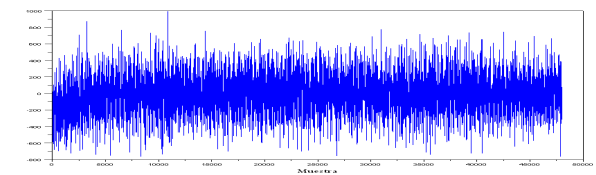
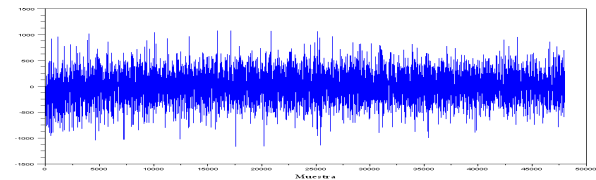
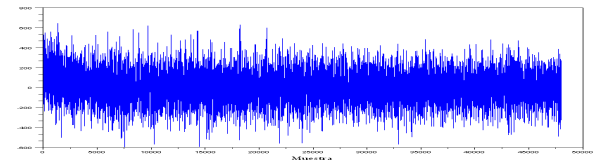
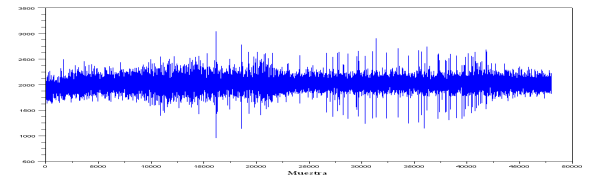
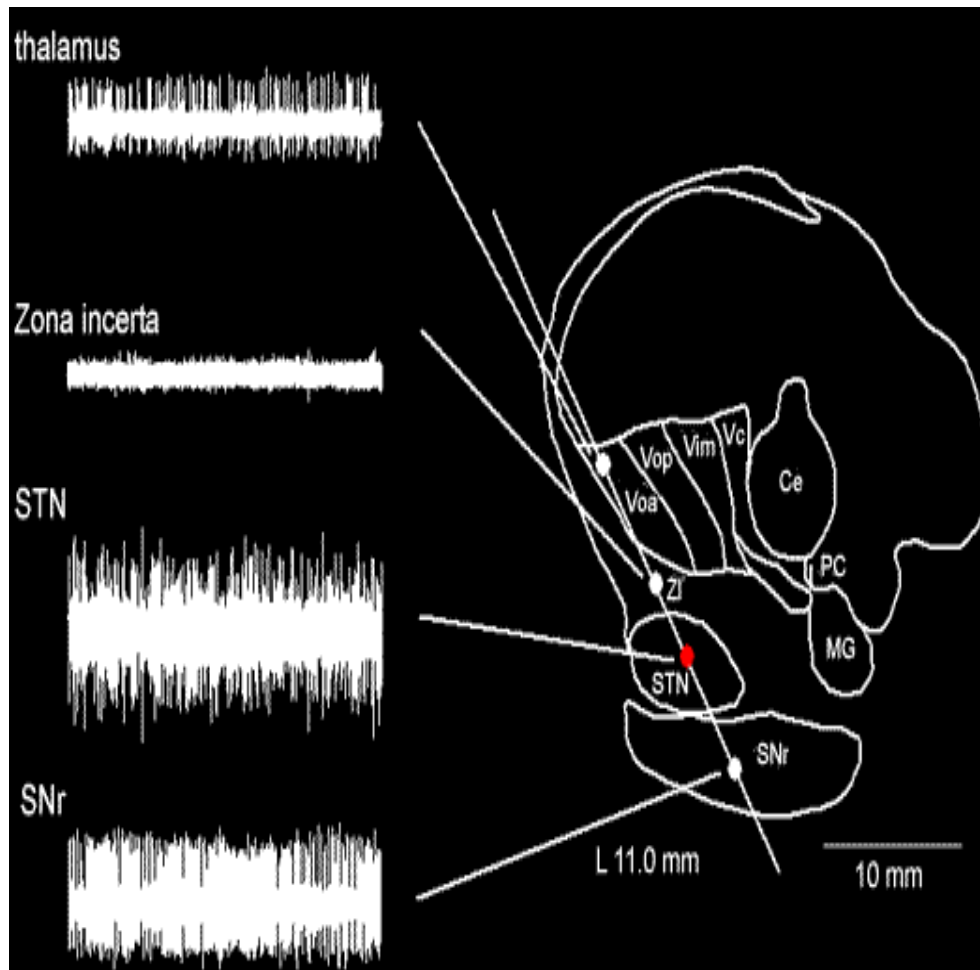
Summary

- Machine learning is the study of how to make computers learn.
- A branch of **artificial intelligence**, concerned with the design and development of algorithms that allow computers to evolve behaviors based on empirical data.
- As intelligence requires knowledge, it is necessary for the computers to acquire knowledge.
- A learning algorithm needs the following elements: **class of tasks, performance metric, and body of experience.**
- The design of a learning algorithm requires to define **the knowledge to learn, the representation of the target knowledge, and the learning mechanism.**
- Machine learning counts with many successful applications and is becoming increasingly important in science and industry.

TIME CODE

- **Parkinson disease (PD) is thought to affect at least 100 persons in every 100,000. The cardinal symptoms of tremor, bradykinesia, postural instability, and rigor result in substantial disability for patients with PD**
- **During the course of the disease, up to 50% of patients will have symptoms refractory to medication and will experience drug-induced dyskinesias**
- **Parkinson's Disease (PD) is primary related to substantia nigra degeneration and, thus, dopamine insufficiency**
- **Disease progression causes L-DOPA therapy efficiency decay (on-off symptom fluctuation), and neurologists often decide to classify patients for DBS (Deep Brain Stimulation) surgery**
- **DBS involves the surgical implantation of electrodes into deep structures of the brain to modulate brain circuitry in an effort to restore normal physiological function**
- **The risk of errors in the localization of the surgical target for deep brain electrical stimulation (DBS) requires the use of some form of intraoperative neurophysiological monitoring to confirm the correct destination during surgery**

Deep brain stimulation



- **Intra-operative microelectrode records were acquired in Parkinsonian patients, awake and unmedicated, subject to deep brain implantation under electrostimulation. Five Parkinsonian patients (4 males and 1 female) aged between 55 ± 6 years old participated voluntarily assuming previously signed consent**
- **The acquisition time for each record was 2 seconds with a sampling frequency of 24 KHz (24,000 samples per second). In total, the database comprises 52 micro recordings, 13 for each of the subcortical structures: Thalamus nucleus (TAL), Zona Incerta (Zi), Subthalamic nucleus (STN) and Substantia Nigra (SNR)**
- **Next, a machine learning algorithm for supervised classification based on deep learning is used with the values of the obtained features, and we show that deep learning can identify and predict with high precision any of the subcortical structures (4 classes): Thalamus (TAL), zona incerta (Zi), subthalamic nucleus (STN) and substantia nigra (SNR)**

Features

$$L = \sum_{i=1}^{N-1} |x_{i+1} - x_i| \quad (6)$$

$$Y = \frac{3}{N-1} \sqrt{\sum_{i=1}^N (x_i - \bar{X})^2} \quad (7)$$

$$\kappa = \frac{1}{2} \sum_{i=1}^{N-2} \max\{0, |\operatorname{sgn}[x_{i+1} - x_i] - \operatorname{sgn}[x_{i+2} - x_{i+1}]]|\} \quad (8)$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^N x_i^2}{N}} \quad (9)$$

$$\Psi = \frac{1}{N-2} \sum_{i=2}^{N-1} x_i^2 - x_{i-1}x_{i+1} \quad (10)$$

$$\kappa = \frac{1}{2} \sum_{i=1}^{N-1} |\operatorname{sgn}(x_{i+1}) - \operatorname{sgn}(x_i)| \quad (11)$$

- As mentioned before when we presented the database, each record was acquired for 2 seconds at a sampling frequency of 24 KHz, which leads to each record having 48,000 samples
- If we consider a trajectory of 13 records for each of the subcortical structures: Thalamus nucleus (TAL), Zona Incerta (Zi), Subthalamic nucleus (STN) and Substantia Nigra (SNR), the final trajectory is made up of 52 records and has a total of 2,496,000 samples
- Next, the final trajectory is divided into windows 4992 consecutive samples and for each of these windows the six features are determined, yielding a total of 500 instances (patterns) by feature. The decomposition described above is presented in matrix form as follows:

$$X = \begin{matrix} & V_1 & V_2 & \dots & \dots & V_p \\ \begin{matrix} X_1 \\ X_2 \\ \vdots \\ \vdots \\ X_n \end{matrix} & \begin{bmatrix} x_{11} & x_{12} & \dots & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \dots & \vdots \\ \vdots & \vdots & \dots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & \dots & x_{np} \end{bmatrix} \end{matrix}$$

- Example 2 to understand machine learning
- Program: Example2ntroML.py

Machine Learning Problems

	<i>Supervised Learning</i>	<i>Unsupervised Learning</i>
<i>Discrete</i>	classification or categorization	clustering
<i>Continuous</i>	regression	dimensionality reduction

What is Prediction?

- (Numerical) prediction is similar to classification
 - construct a model
 - use model to predict continuous or ordered value for a given input
- Prediction is different from classification
 - Classification refers to predict categorical class label
 - Prediction models continuous-valued functions
- Major method for prediction: regression
 - model the relationship between one or more *independent* or **predictor** variables and a *dependent* or **response** variable
- Regression analysis
 - Linear and multiple regression
 - Non-linear regression
 - Other regression methods: generalized linear model, Poisson regression, log-linear models, regression trees, etc

Linear Regression

- Linear regression: involves a response variable y and a single predictor variable x

$$y = w_0 + w_1 x$$

where w_0 (y-intercept) and w_1 (slope) are regression coefficients

- Method of least squares: estimates the best-fitting straight line

$$w_1 = \frac{\sum_{i=1}^{|D|} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|} (x_i - \bar{x})^2} \quad w_0 = \bar{y} - w_1 \bar{x}$$

- Multiple linear regression: involves more than one predictor variable
 - Training data is of the form $(\mathbf{X}_1, y_1), (\mathbf{X}_2, y_2), \dots, (\mathbf{X}_{|D|}, y_{|D|})$
 - Ex. For 2-D data, we may have: $y = w_0 + w_1 x_1 + w_2 x_2$
 - Many nonlinear functions can be transformed into the above

Nonlinear Regression

- Some nonlinear models can be modeled by a polynomial function
- A polynomial regression model can be transformed into linear regression model. For example,

$$y = w_0 + w_1 x + w_2 x^2 + w_3 x^3$$

convertible to linear with new variables: $x_2 = x^2$, $x_3 = x^3$

$$y = w_0 + w_1 x + w_2 x_2 + w_3 x_3$$

- Other functions, such as power function, can also be transformed to linear model

Other Regression-Based Models

- Generalized linear model:
 - Foundation on which linear regression can be applied to modeling categorical response variables
 - Logistic regression: models the prob. of some event occurring as a linear function of a set of predictor variables
- Log-linear models: (for categorical data)
 - Approximate discrete multidimensional prob. distributions
 - Also useful for data compression and smoothing
- Regression trees and model trees
 - Trees to predict continuous values rather than class labels

Instance based learning

- Approximating real valued or discrete-valued target functions
- Learning in this algorithm consists of storing the presented training data
- When a new query instance is encountered, a set of similar related instances is retrieved from memory and used to classify the new query instance
- Construct only local approximation to the target function that applies in the neighborhood of the new query instance
- Never construct an approximation designed to perform well over the entire instance space
- Appropriate definition of “neighboring” instances
- Disadvantage of instance-based methods is that the costs of classifying new instances can be high
- Nearly all computation takes place at classification time rather than learning time

K-Nearest Neighbor algorithm

- Most basic instance-based method
- Data are represented in a vector space
- Supervised learning

Feature space

$$\left\{ \langle \vec{x}^{(1)}, f(\vec{x}^{(1)}) \rangle, \langle \vec{x}^{(2)}, f(\vec{x}^{(2)}) \rangle, \dots, \langle \vec{x}^{(n)}, f(\vec{x}^{(n)}) \rangle \right\}$$

$$\vec{x} = \begin{cases} x_1 \\ x_2 \\ \dots \\ x_d \end{cases} \in \Re^d \quad \|\vec{x} - \vec{y}\| = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$$

K-Nearest Neighbor algorithm

- In nearest-neighbor learning the target function may be either discrete-valued or real valued
- Learning a discrete valued function
- $f : \mathbb{R}^d \rightarrow V$, V is the finite set $\{v_1, \dots, v_n\}$
- For discrete-valued, the k -NN returns the most common value among the k training examples nearest to xq .

K-Nearest Neighbor algorithm

- Training algorithm
 - For each training example $\langle x, f(x) \rangle$ add the example to the list
- Classification algorithm
 - Given a query instance x_q to be classified
 - Let $x_1, \dots, x_k$ k instances which are nearest to x_q

$$\hat{f}(x_q) \leftarrow \arg \max_{v \in V} \sum_{i=1}^k \delta(v, f(x_i))$$

- Where $\delta(a, b) = 1$ if $a = b$, else $\delta(a, b) = 0$ (Kronecker function)

How to determine the good value for k ?

- Determined experimentally
- Start with $k=1$ and use a test set to validate the error rate of the classifier
- Repeat with $k=k+2$
- Choose the value of k for which the error rate is minimum
- Note: k should be odd number to avoid ties

Logistic Regression

- Form of regression that allows the prediction of discrete variables by a mix of continuous and discrete predictors.
- Addresses the same questions that discriminant function analysis and multiple regression do but with no distributional assumptions on the predictors (the predictors do not have to be normally distributed, linearly related or have equal variance in each group)
- Logistic regression is often used because the relationship between the DV (a discrete variable) and a predictor is non-linear

π = Proportion of “Success”

In ordinary regression the model predicts the *mean* Y for any combination of predictors.

What’s the “mean” of a 0/1 indicator variable?

$$\bar{y} = \frac{\sum y_i}{n} = \frac{\# \text{ of } 1's}{\# \text{ of trials}} = \text{Proportion of "success"}$$

Goal of logistic regression: Predict the “true” proportion of success, π , at any value of the predictor.

The logistic function

$$\hat{Y}_i = \frac{e^u}{1 + e^u}$$

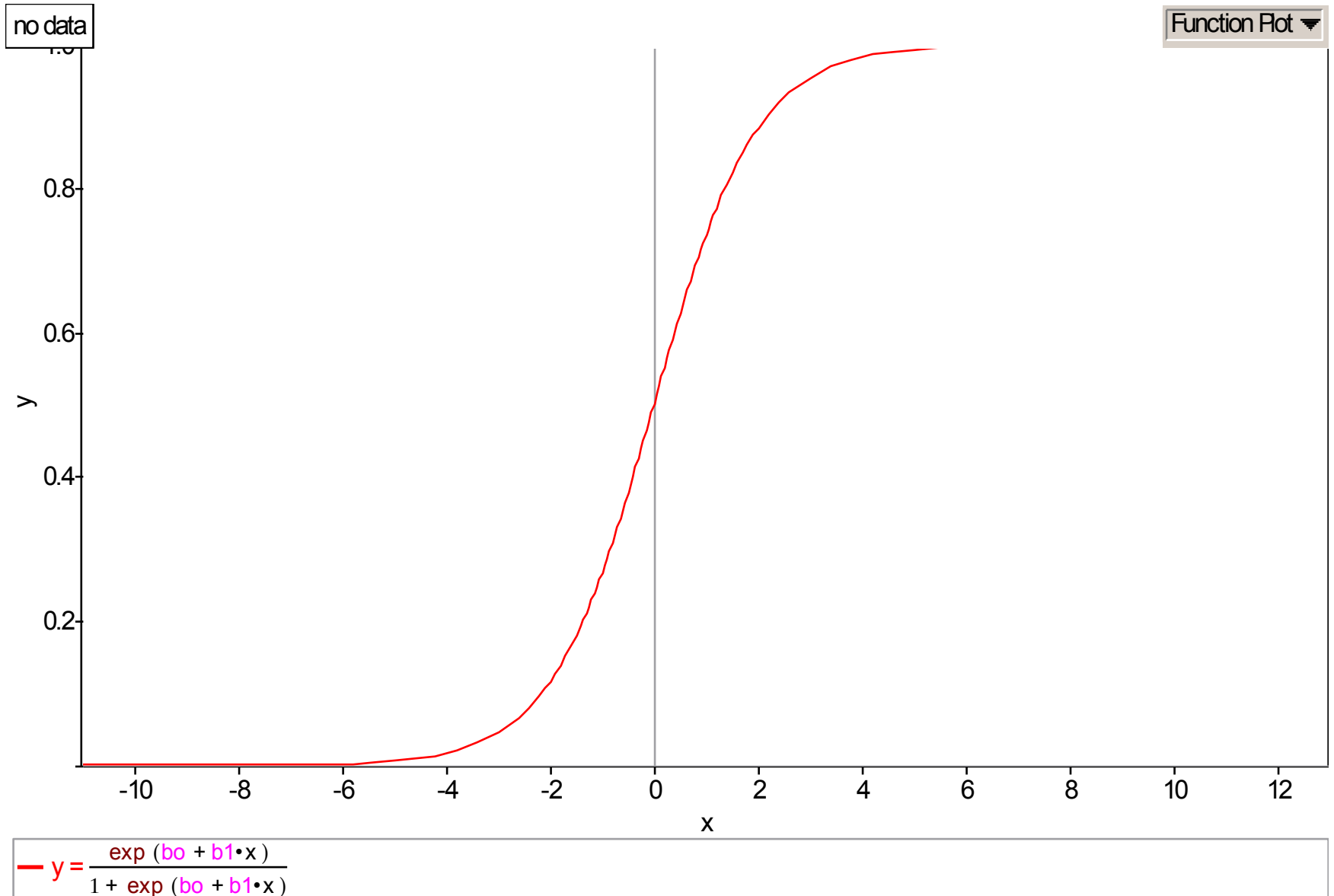
- Where $\hat{Y}$ is the estimated probability that the i -th case is in a category and u is the regular linear regression equation:

$$u = A + B_1X_1 + B_2X_2 + \cdots + B_KX_K$$

The logistic function

$$\hat{\pi}_i = \frac{e^{b_0 + b_1 X_1}}{1 + e^{b_0 + b_1 X_1}}$$

Logit Function



Discriminant Analysis

Classification is an important component of all scientific research. Statistical techniques concerned with classification are essentially of two types.

The first ([cluster analysis](#)) aims to uncover groups of observations from initially unclassified data.

The second (**discriminant analysis**) works with data that is already classified into groups to derive rules for classifying new (and as yet unclassified) individuals on the basis of their observed variable values.

Discriminant analysis is used to determine which variables discriminate between two or more naturally occurring groups.

Linear Discriminant Functions

A discriminant function is a linear combination of the components of x :

$$g(x) = w^t x + w_0$$

where

- w^t is the weight vector
- w_0 is the bias or threshold weight

For the two-class problem we can use the following decision rule:

Decide c_1 if $g(x) > 0$ and c_2 if $g(x) < 0$.

For the general case we will have one discriminant function for each class.

The Problem with Multiple Classes

How do we use a linear discriminant when we have more than two classes?

There are two approaches:

1. Learn one discriminant function for each class
2. Learn a discriminant function for all pairs of classes

If c is the number of classes, in the first case we have c functions and in the second case we have $c(c-1) / 2$ functions.

In both cases we are left with ambiguous regions.

Figure 5.3

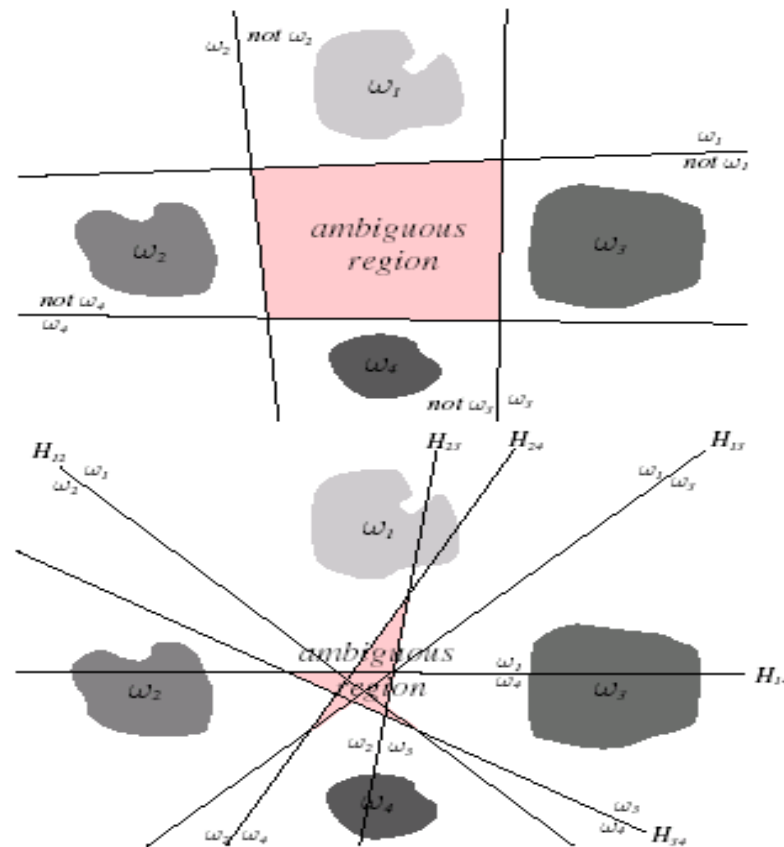


FIGURE 5.3. Linear decision boundaries for a four-class problem. The top figure shows ω_i / not ω_i dichotomies while the bottom figure shows ω_i / ω_j dichotomies and the corresponding decision boundaries H_{ij} . The pink regions have ambiguous category assignments. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Linear Machines

To avoid the problem of ambiguous regions we can use linear machines:

- We define c linear discriminant functions and choose the one with highest value for a given x .

$$g_k(x) = w_k^T x + w_{k0} \quad k = 1, \dots, c$$

- In this case the decision regions are convex (a set which contains the entire line segment joining any pair of its points) and thus are limited in flexibility and accuracy.

Figure 5.4

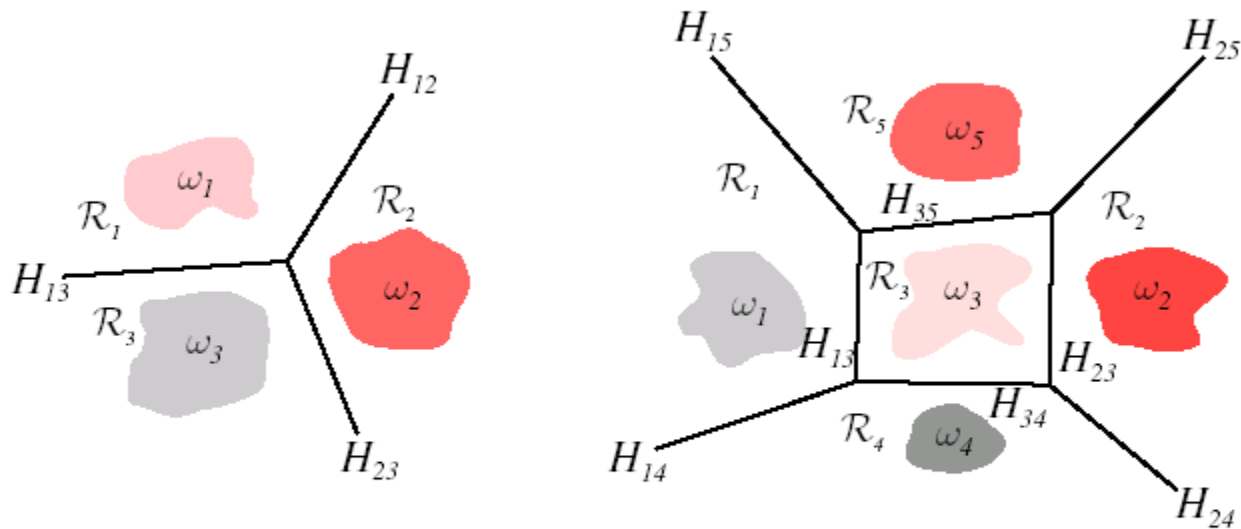


FIGURE 5.4. Decision boundaries produced by a linear machine for a three-class problem and a five-class problem. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

Support Vector Machines

- Support Vector Machines (SVMs) were invented by Vladimir Vapnik for classification based on prior knowledge [Vapnik98, Burges98].
- Create classification functions from a set of labeled training data.
 - The output is binary: is the input in a category?
- SVMs find a hypersurface in the space of possible inputs.
 - Split the positive examples from the negative examples.
 - The split is chosen to have the largest distance from the hypersurface to the nearest of the positive and negative examples.
- SVMs can be used to multiclass classification

Estimating Error

- Cross-validation (k -fold, where $k = 10$ is most popular)
 - Randomly partition the data into k *mutually exclusive* subsets, each approximately equal size
 - At i -th iteration, use D_i as test set and others as training set
 - Stratified cross-validation: folds are stratified so that class dist. in each fold is approx. the same as that in the initial data

Metrics for Performance Evaluation

- Focus on the predictive capability of a model
 - Rather than how fast it takes to classify or build models, scalability, etc.
- Confusion Matrix:
 - The diagonal elements show the number of correct classifications for each class
 - The off-diagonal elements provides the misclassifications

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	a	b
	Class=No	c	d

a: TP (true positive)
b: FN (false negative)
c: FP (false positive)
d: TN (true negative)

Metrics for Performance Evaluation

ACTUAL CLASS	PREDICTED CLASS	
	Class=Yes	Class=No
Class=Yes	a (TP)	b (FN)
	c (FP)	d (TN)

- Most widely-used metric:

$$\text{Accuracy} = \frac{a + d}{a + b + c + d} = \frac{TP + TN}{TP + TN + FP + FN}$$

More measures

- True Positive Rate (TPR) (Sensitivity)
 - $a/a+b$
- True Negative Rate (TNR) (Specificity)
 - $d/c+d$
- False Positive Rate (FPR)
 - $c/c+d$
- False Negative Rate (FNR)
 - $b/a+b$

Predictor Error Measures

- Measure predictor accuracy: measure how far off the predicted value is from the actual known value
- **Loss function:** measures the error betw. y_i and the predicted value y_i'
 - Absolute error: $|y_i - y_i'|$
 - Squared error: $(y_i - y_i')^2$
- Test error (generalization error): the average loss over the test set
 - Mean absolute error: $\frac{\sum_{i=1}^d |y_i - y_i'|}{d}$ Mean squared error: $\frac{\sum_{i=1}^d (y_i - y_i')^2}{d}$
 - Relative absolute error: $\frac{\sum_{i=1}^d |y_i - y_i'|}{\sum_{i=1}^d |y_i - \bar{y}|}$ Relative squared error: $\frac{\sum_{i=1}^d (y_i - y_i')^2}{\sum_{i=1}^d (y_i - \bar{y})^2}$

Sklearn.metrics

Compute precision, recall, F-measure and support for each class

- The precision is the ratio $tp / (tp + fp)$ where tp is the number of true positives and fp the number of false positives. The precision is intuitively the ability of the classifier.
- The recall is the ratio $tp / (tp + fn)$ where tp is the number of true positives and fn the number of false negatives. The recall is intuitively the ability of the classifier to find all the positive samples.
- The F-beta score can be interpreted as a weighted harmonic mean of the precision and recall, where an F-beta score reaches its best value at 1 and worst score at 0.
- The F-beta score weights recall more than precision by a factor of β .
 $\beta = 1.0$ means recall and precision are equally important.
- The support is the number of occurrences of each class in y_true .

TIME CODE

HOMEWORK 1

- Use the Program `Example2ntroML.py` for the dataset: `data11tumors.csv`, and evaluate the performance for each method.